ENGR 1451/2451: Exploratory Data Science

Two-Sample t-Tests

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Today: Comparing Two Averages

From one sample to two

One sample versus a fixed benchmark:

$$\bar{x} - \mu_0.$$

Two independent samples:

$$\bar{x}_A - \bar{x}_B.$$

- Find the SE of a difference.
- Test a difference in population means.
- Distinguish independent groups from paired data.
- Connect the statistical result to the study design.

With two independent groups, both sample averages contribute uncertainty.

New Heat Treatment: Experimental vs. Control

A lab randomly assigns **24 comparable steel coupons** to the **new heat-treatment cycle** (treatment) or the **standard cycle** (control), with 12 coupons per group. Vickers hardness is measured once per coupon.

Group	n	Mean hardness (HV)	Sample SD (HV)
New heat treatment (T)	12	330	18
Standard cycle (C)	12	314	14

Question

Does the new heat treatment **increase the population mean hardness**?

The observed difference is $330 - 314 = 16$ HV. Is that convincing evidence of an increase?

A Fixed Target or a Control Group?

One-sample t-test

Suppose the design target is **320 HV**.

$$H_0 : \mu_T = 320, \qquad H_A : \mu_T > 320.$$

Observed discrepancy: $330 - 320 = 10$ HV.
Only the treated mean has sampling uncertainty.

Two-sample t-test

Compare the treated and control groups:

$$\Delta = \mu_T - \mu_C, \quad \widehat{\Delta} = 330 - 314 = 16.$$

$$H_0 : \Delta = 0, \qquad H_A : \Delta > 0.$$

Both group means contribute sampling uncertainty.

A fixed target and a measured control group answer different questions.

Standard Error of a Difference

For **independent groups**, the variances of the sample means add:

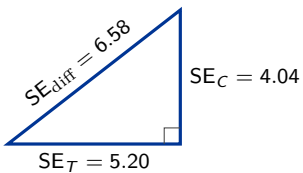
$$\widehat{\mathrm{SE}}(\bar{X}_T - \bar{X}_C) = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}$$

Heat-treatment example:

$$\widehat{\mathrm{SE}} = \sqrt{\frac{18^2}{12} + \frac{14^2}{12}} \approx 6.58 \text{ HV}.$$

Why is the SE of the difference larger?

Each mean varies from sample to sample. Both sources of uncertainty matter, even when we *subtract* the averages.



Two-Sample t-Test: The Main Idea

Standardize the discrepancy (from w09a)

$$\text{test statistic} = \frac{\text{observed difference} - \text{difference under } H_0}{\text{estimated SE of the difference}}$$

For two **independent groups**, where Δ_0 is the difference claimed by H_0:

$$\boxed{t = \frac{(\bar{x}_A - \bar{x}_B) - \Delta_0}{\sqrt{s_A^2/n_A + s_B^2/n_B}}} \qquad (\text{often } \Delta_0 = 0).$$

Check the data and design

Independent observations; no strong outliers; approximately normal group distributions are especially important for small samples.

Use Welch's reference

Software uses a t distribution with approximate degrees of freedom. Equal population variances are *not* assumed.

Heat-Treatment Result: Is Hardness Higher?

Treated minus standard control

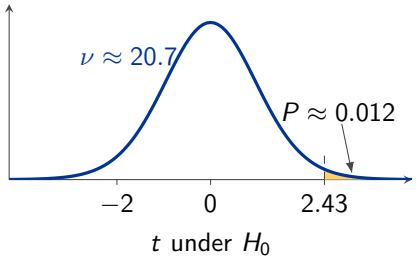
$$H_0 : \Delta = 0, \qquad H_A : \Delta > 0.$$

$$t = \frac{(330 - 314) - 0}{6.5828} \approx 2.43.$$

Software reports Welch $df \approx 20.7$.

$$P = \Pr(T_\nu \geq 2.43) \approx 0.0121.$$

At $\alpha = 0.05$, should we reject H_0?

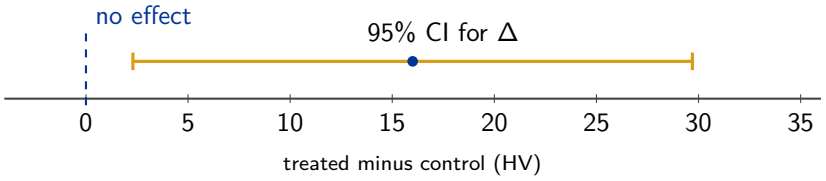


Confidence Interval for the Treatment Effect

A **two-sided 95% interval** estimates $\Delta = \mu_T - \mu_C$:

$$\widehat{\Delta} \;\pm\; t^{*}\widehat{\mathrm{SE}}, \qquad t^{*} = t_{0.975,\nu}.$$

$$16 \pm 2.081(6.583) \;\approx\; [2.30,\; 29.70] \text{ HV}.$$



Course extension connecting two-sample intervals to w07b and w09a. Hypothetical data.

Python: Independent Samples from Summary Statistics

When each group is summarized by n, a sample mean, and a **sample SD**:

```
from scipy import stats

result = stats.ttest_ind_from_stats(
    mean1=330, std1=18, nobs1=12,  # treatment
    mean2=314, std2=14, nobs2=12,  # control
    equal_var=False,
    alternative="greater"
)
print(result.statistic, result.pvalue)
# t = 2.4306, P = 0.0121
```

Key options

`equal_var=False` selects **Welch's test**; equal population variances are not assumed.

Direction and order

`"greater"` tests group 1 > group 2. Other choices are `"less"` and `"two-sided"`.

Experiments: Vitamin C versus Placebo

200 subjects are **randomly assigned** to vitamin C or placebo:

Group	n	Average number of colds	SD (book)
Vitamin C	100	2.3	3.1
Placebo	100	2.6	2.9

Observed difference (treatment minus control): $2.3 - 2.6 = -0.3$ colds. The approximate one-sided P-value is 0.24.

Interpretation check

Why is random assignment useful? Does $P = 0.24$ prove vitamin C has no effect?

Does a Study App Improve Scores?

Fifty students are **randomly assigned** to a study app or a control condition.

- Average score with app: 82
- Average score without app: 77
- One-sided two-sample t-test: $P \approx 0.066$

Discussion

1. State H_0 and H_A.
2. At $\alpha = 0.05$, what conclusion can we draw?
3. Does the result show that the app does *not* work?

Paired Data: Compare Within Each Pair

Age-adjusted vocabulary scores from **400 randomly sampled two-child families**:

Two measurements per family

	Mean	SD
First-born	29	10
Second-born	28	10

The average paired difference is $\bar{d} = 29 - 28 = 1$ point.

Analyze one sample of differences

$$d_i = x_{\text{first},i} - x_{\text{second},i}.$$

For $H_0 : \mu_d = 0$,

$$\boxed{t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad df = n - 1.}$$

Here $n = 400$ **pairs**, not 800 independent children.

Can we finish the test from the two means and two SDs?

Design Check: Which Analysis?

First decide whether the comparison uses a **fixed target**, **independent groups**, or **paired observations**.

Study	Appropriate analysis
Treated hardness versus a fixed 320 HV specification.	
Different coupons assigned to the new or standard heat-treatment cycle.	
The same 20 coupons measured before and after treatment.	
Independent samples from two different school years.	

Pairing is a feature of the data collection, not something chosen after seeing the data.

Summary of Mean Tests

Test	When to use it	Python (SciPy)
One-sample z	One mean vs. μ_0; σ known.	Compute z; use `stats.norm.cdf(z)` or `stats.norm.sf(z)`.
One-sample t	One mean vs. μ_0; population SD unknown.	`stats.ttest_1samp(x, popmean=mu0)`
Two-sample z	Two independent means; population SDs known.	Compute z; use `stats.norm.cdf(z)` or `stats.norm.sf(z)`.
Two-sample t	Two independent means; population SDs unknown.	`stats.ttest_ind(a, b, equal_var=False)`
Paired t	Before/after or matched observations; analyze the within-pair differences.	`stats.ttest_rel(after, before)`

Summary: Compare the Right Difference

1. **Identify the design:** independent or paired observations; note whether treatment was randomly assigned
2. **Define the population difference** and choose the alternative in advance.
3. **Use the appropriate SE:**

$$\text{Independent: } \sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}} \qquad \text{Paired: } \frac{s_d}{\sqrt{n}}.$$

4. **Report the difference, uncertainty, and P-value** in context.

Keep these conclusions separate

Statistical significance, practical importance, and causation are different questions.